AI NOTES SERIES

Artificial Intelligence

The Field as a Whole — what AI really is, how its branches fit together, and what it is not.

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These are introductory study notes, written in simple English for teachers, students and beginners. They come from my own learning and exploration of the subject. Explanations are simplified on purpose, so the main ideas are easy to follow and easy to revise.

The Big Idea

Artificial intelligence, usually shortened to **AI**, is a field of computer science. Its goal is to build computer systems that can do tasks which normally need human intelligence — tasks such as understanding language, recognising images, learning from examples, and making simple decisions.

The most important point in these notes is simple: **AI is not a single technology.** It is not one app, one website, or one machine. AI is an umbrella field — one big field that holds many different methods, tools and branches under it, all working towards that same goal.

> **Plain definition:** Artificial intelligence is the field of making computers able to perform tasks that seem to need intelligence, by learning patterns from data or by following carefully designed rules.

Why the umbrella picture helps

- When people say "AI" in the news, they may mean very different things — a translation app, a photo tool, or a chatbot. All of them sit under the same umbrella, but they are not the same thing.
- Some AI systems follow rules written by people. Most modern AI systems learn from examples instead. Both belong to the same field.
- Learning the map of the field first makes every later topic easier, because you can see where each new idea belongs.

A simple everyday picture

Think of a school. "Education" is the umbrella word, but inside it there are different subjects — language, mathematics, science — and different classes inside each subject. AI works the same way. The field is the school, the branches are the subjects, and the apps and tools you use are like individual lessons.

Keep this picture in mind as you read the next chapter, which draws the map of the whole field.

The Map of the Field

The branches of AI are connected like a tree. At the top is the whole field. Inside it, one approach does most of the modern work, and the well-known branches grow out of that approach.

Artificial Intelligence — the whole field

- Machine Learning — the main modern approach: systems learn patterns from data instead of following only fixed rules
  - Deep Learning — machine learning using large neural networks with many layers
    - Natural Language Processing — working with human language
    - Computer Vision — understanding images and video
    - Generative AI — creating new text, images, audio and video
    - Multimodal AI — working with several kinds of input together, such as text and images
    - AI Agents — systems that take steps and use tools to complete a task

At the centre of much current progress: large language models, built on the transformer design.

How to read the map

- **Machine learning** is the engine of modern AI. Instead of a programmer writing every rule, the system is shown many examples and finds patterns in them.
- **Deep learning** is a powerful kind of machine learning. It uses neural networks with many layers, which lets it handle difficult material such as speech, photos and long pieces of text.
- The branches below deep learning in the map are areas of work built mostly on deep learning methods. One system can belong to several branches at once — a tool that looks at a photo and writes about it is both computer vision and multimodal AI.

Language models and the transformer

A **large language model** is a model trained on very large amounts of text. It learns patterns in language and can then write, answer, summarise and translate. Most large language models today are built on a design called the **transformer**. In simple words, the transformer is good at paying attention to the parts of a sentence that matter most for meaning, which is why modern language tools improved so quickly.

The Fuel: Data

Every branch on the map depends on **data** — information such as text, images, sound, numbers or video. Data is the fuel of AI in the same way that petrol is the fuel of a car. Without it, the engine of machine learning cannot run.

How learning from data works, in simple steps

- A system is given a very large number of examples.
- It looks for patterns in those examples — for instance, what words usually come together, or what shapes usually appear in photos of cats.
- Those patterns are stored inside a **model**, which is the trained result that can then work on new examples it has never seen.
- People check the model's answers and use more or better data to improve it.

Patterns, not understanding

This is the key idea of the chapter. An AI system finds **patterns** — regular connections in data. Finding a pattern is not the same as understanding. A model can learn that certain words often appear near each other without knowing what any of them truly mean, in the way a person knows.

> **Remember:** Good data leads to better patterns. Weak, wrong or unfair data leads to weak, wrong or unfair results. This is why people who work with AI care so much about where data comes from.

What this means in daily life

- An AI tool is usually strongest on topics and languages that appear often in its training data, and weaker where data is rare.
- An answer can sound confident and still be wrong, because the model is following patterns, not checking facts like a person would.
- Personal data deserves care. What we share with AI tools can become part of someone's data, so share thoughtfully.

Narrow AI and Beyond

AI systems are often sorted by how wide their ability is. Only the first of these kinds exists today.

Narrow AI — the AI we actually have

Narrow AI is built for a limited task or a small set of tasks. A translation tool translates. A chess program plays chess. A photo tool recognises faces in pictures. Each one can be very good at its own job, but it cannot step outside that job and think about everything else. Every AI system in use today — including the most advanced chatbots and image tools — is narrow AI.

General AI — an idea, not a product

General AI would be a system that could learn and handle almost any intellectual task a human can, across many different areas, the way a person can move from cooking to mathematics to learning a new language. General AI **does not exist yet**. It is a goal and a subject of research and debate, not something anyone can buy or use today.

Superintelligence — a future idea

Superintelligence is the idea of a system far more capable than humans in almost every area. It appears in books, films and serious discussions about the future, but it also **does not exist**. When you meet these words in the news, they are almost always about the future, not about a tool that exists now.

> **Simple test:** If a product exists and people can use it today, it is narrow AI — however impressive it feels. General AI and superintelligence are names for ideas about what might come later.

What AI Is Not

Because AI tools can write smoothly and answer quickly, it is easy to imagine there is a mind inside them. Being clear about what AI is not protects us from that mistake.

- **AI is not alive.** It is software running on computers. It does not grow, feel hunger or tiredness, or have a life of its own.
- **AI is not conscious.** There is no inner experience, no feelings, and no awareness behind its answers.
- **AI does not understand like a human.** It works with patterns in data. A person who reads a story knows the people in it are happy or sad; a model works out which words usually follow which other words.
- **Capability is not comprehension.** A system can be able to do a task very well — translate, summarise, calculate — without understanding the subject the way a student who studied it would.

None of this makes AI useless. A calculator does not understand mathematics either, and it is still a valuable tool. The point is to use AI as a tool, with human judgement in charge.

Where the Field Is Used

- **Search** — finding the most useful pages and answers for a question.
- **Translation** — turning text and speech from one language into another.
- **Accessibility** — captions for people who cannot hear well, and text read aloud for people who cannot see well.
- **Health support** — helping professionals spot patterns in scans and records; final decisions stay with qualified people.
- **Education** — practice questions, explanations at a student's level, and help for teachers preparing lessons.
- **Science** — sorting very large amounts of research data and suggesting patterns for scientists to test.

Words to Know

Artificial intelligence — the field of building computer systems that can do tasks which normally need human intelligence, such as understanding language or recognising images.

Machine learning — a way of building AI in which a system learns patterns from examples and data, instead of following only rules written by people.

Deep learning — a powerful kind of machine learning that uses neural networks with many layers to handle difficult material such as speech, photos and text.

Neural network — a computing system loosely inspired by the network of cells in a brain, made of connected units that pass signals to each other and adjust their strength during training.

Model — the trained result of machine learning: the stored patterns a system uses to answer questions or make predictions about new material.

Data — information used to train or feed an AI system, such as text, images, sound, video or numbers.

Algorithm — a clear, step-by-step set of instructions for solving a problem or completing a task, which a computer can follow.

Transformer — a model design that pays attention to the most important parts of its input, such as the key words in a sentence; it sits at the centre of most modern language models.

Multimodal — able to work with more than one kind of input or output together, for example text with images, or speech with video.

Agent — an AI system that can take a series of steps, and sometimes use tools, to complete a task with less step-by-step guidance from a person.

Quick Revision

- AI is one umbrella field, not a single technology, app or machine.
- Machine learning is the main modern approach: systems learn patterns from data instead of following only fixed rules.
- Deep learning is machine learning with large, many-layered neural networks, and most well-known branches grow from it.
- Large language models, built on the transformer design, sit at the centre of much current progress.
- Data is the fuel of every branch. Better data gives better patterns; poor data gives poor results.
- AI finds patterns — it does not understand the world the way people do.
- All AI that exists today is narrow AI. General AI and superintelligence do not exist yet.
- AI is not alive and not conscious. Capability is not comprehension.
- The field is used in search, translation, accessibility, health support, education and science — always with human judgement in charge.

Closing Note

These notes are an introduction to the field as a whole. Explanations are simplified on purpose, so a beginner can build a clear picture first and add detail later. They are study notes from my own learning and exploration, shared for students, teachers and anyone starting their journey with artificial intelligence.

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